Computer Engineer

Natural Sciences
(Scire – To Know)

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Chair

Outline the history of Science from Ancient Greece (400BC – 200 AD) to Present so that

- You will get a sense of how science evolved
- Your interest in science and engineering will increase
- We would like to you to see the big picture

Outline

- Discuss

About the Course

- Attendance is expected..
- Grading based on
  - Midterm
  - Papers / Presentations /
  - Final

Cave Man

- Wondered why mountains rumbled during a storm

- They practiced
  - Science when hypothesized god got angry
  - Political Science: Gave offerings to god
  - Religion: They worship

Ancient Greeks – Birthplace of Olympics

Social Sciences

- Democracy, Freedom
- Philosophy - Humanism
- Play – Comedy, Tragedy…

A systematic Approach to Natural Sciences

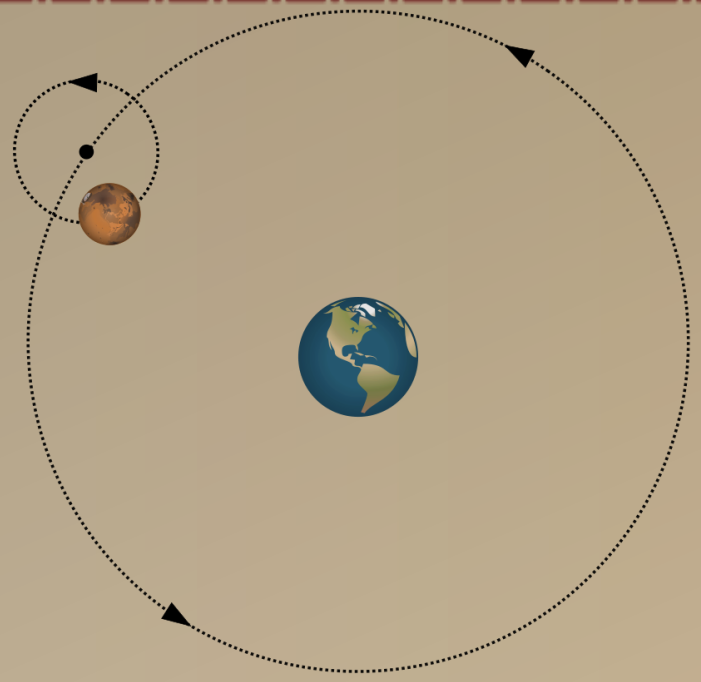
Observe the nature and explain – cause and effect (Aristotle)

Idea of Prediction of nature by universal laws

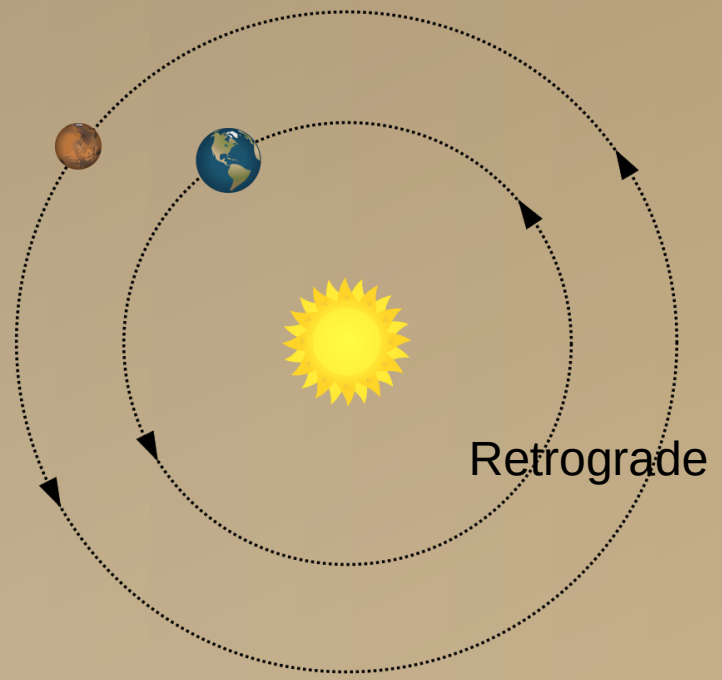
- Before Greeks mainly sprites

Use math to prove – Euclid, Pythagorean theorem, Archimedes – Traces of Engineering

Ancient Greeks - Astronomy



geocentric system



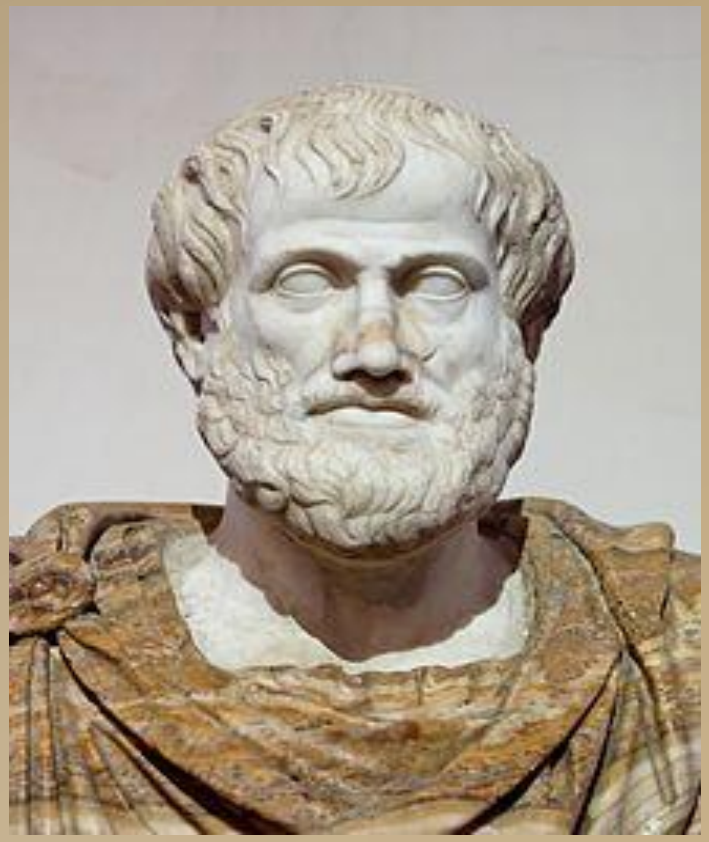
heliocentric system

Astronomy

- Planets move in complex paths – Retrograde
- Geocentric model of Ptolemy

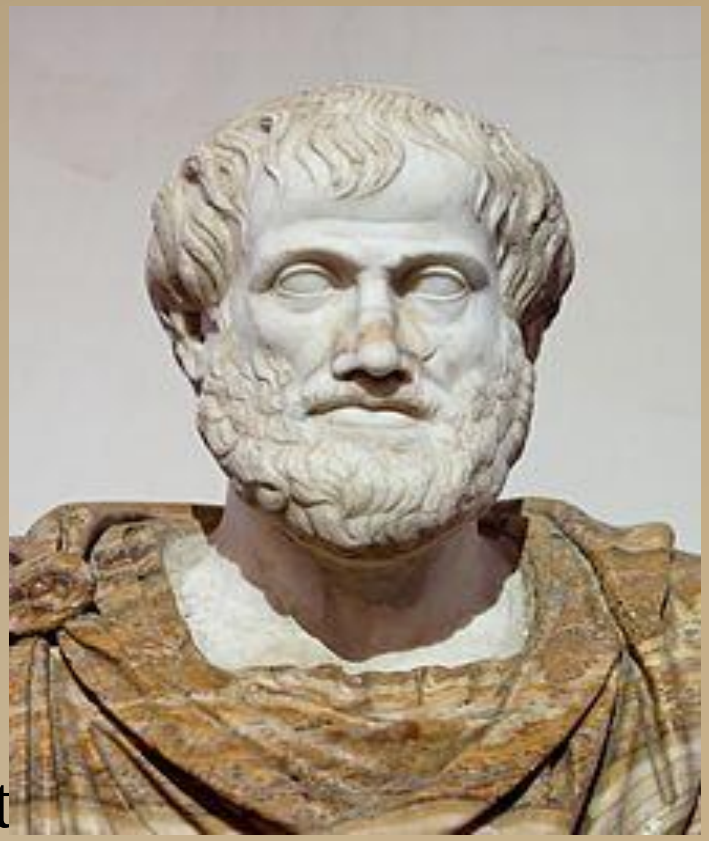
First Scientist Aristotle ~ 300 BC

- Student of Plato – Tutor of Alexander the Great. (spread of knowledge)
  - Plato is a Philosopher, Laid the foundations of western philosophy, established the first academy



First Scientist Aristotle ~ 300 BC

- Aristotle radically transformed most, if not all, areas of knowledge he touched
  - Classified areas of human knowledge into distinct disciplines such as mathematics, biology, and ethics
  - Classification of living things
  - Nature is composed of things that change
  - Earth is round and center of Universe



Aristotle Philosophy

The universe is made up of 4 elements:

Earth, water, **air** and fire. Each seek their own level. Solid sinks, Rain Falls, Air bubbles. Flame rises.

There is change and it arises from 4 causes. Material, Formal(Shape), Efficient(Doer/Maker), Final (purpose in nature)

Ex: Wood Table

3 kinds of Movement

1. Qualitative - change in state (Aging, Tree)
2. Quantitative – increase/decrease, (Gain Weight)
3. Change of Location (Regular Movement)

Concept of AIR

Why is it hard ?

Aristotle's Formal Logic

Resolution of disagreement through rational discussion and search for truth

- Deduction would follow from premises
  - Summer is hot
  - Today is summer
  - Today is hot

Aristotle's Formal Logic

- Show that a given hypothesis leads to a contradiction; thus, forcing the withdrawal of the hypothesis
- Denying a presupposition of the contending thesis and antithesis
  - Have you talked to Hans?
    - Presupposition: Hans exists.

BlakeRottinger.com

Bruno

Giordano Bruno (Italian) extended the then novel Copernican model and proposed that the stars were just distant suns surrounded by their own planets and raised the possibility that these planets could even foster life of their own (a philosophical position known as cosmic pluralism).

He also insisted that the universe is in fact infinite and could have no celestial body at its "center".

Galileo – (1564 - 1642)

Physics, Astronomy, Engineer, Mathematician

- Invented
  - Jupiter Moons, Sunspots
  - Compass, and other instruments
  - Kinematics, Strength of Materials

- Arrested for contradicting bible for geocentrism

Between 1700 - 1900

Newton invented Calculus, but it took 100 years to be established by notable scientist (Thomas Jefferson, Voltaire)

Wohler disproved the vitalist theory of life by synthesizing an organic compound – 1828

Darwinism is about 150 years old and still the target of fundamentalist

Einstein established quantum mechanics and relativity. Time is a function of your frame of reference. Famous Equation → Nuclear ERA

Move through Science

Touch major historical achievements

- Basic Principles of Physics, Astronomy, Chemistry, Biology
- Mathematics including arithmetic, euclidean geometry, formal logic, calculus, statistics, boolean algebra
- Social Sciences if time permits
- And 21st Century

Science

Tries to find rational answers to questions

Used to understand and predict

Hypothesis, theory, and law through scientific studies.

Science is independent of culture and time

It is cumulative and progressive

<u>Science is the mathematical description of nature</u>

!

Before the Greeks

Although applied science and mathematics is practiced, until Greeks no coherent theory of science is proposed.

Tally sticks used for counting – 30000 BC. So does accounting and writing

Basic arithmetic is about addition. Subtraction is reverse addition; multiplication is a series of additions and … division is _____.

Agriculture

Babylonians, Egyptians...practiced astronomy and engineering for survival

- Agricultural timeline which requires defining time, seasons.
- Enabled settlements
- Commerce
- Geometry for ownership of land (area)

With geometry, we can design, construct buildings, irrigation system.

It is the foundation of mechanical engineering

Ancient History

https://www.youtube.com/watch?v=kRow4ELfGO0

Mining

Materials for weapons and tools invented – Stone Age

Copper and Tin mined and heated together for an alloy → Bronze Age

Iron obtained by heating
Iron Oxide + Carbon → Carbon Oxide + IRON

Iron was much stronger of any material found. It was a ceremonial element.
Played a key role in the industrial age once machined.

Ancient Science

Some botany and chemistry developed for Agriculture

Herbal Medicine + Incantations healed the patient.

Midwifery practices

Egyptians excelled in advanced geometry and applied engineering. (Pyramids)

Babylonian mathematics was quite advanced – First, Second degree equations. Basic Algebra

Indians advanced in Math, invented zero.

Sumerian's invented writing.

Numbering System for Accounting

Cave Man

http://youtu.be/xSi0lfl-31U

Romans

https://www.youtube.com/watch?v=15sLPp0YvQk

Computation

https://www.youtube.com/watch?v=5sS7w-CMHkU

Who invented the Zero ?

- Following in the same lines, Zero as an entity means nothing, as we already know now.
  - Sumerian (~ 4000 BC) - Empty as Placeholder
- But the early concept of giving a symbol to nothing would have been considered abstract then.
  - Babylonians (~ 1000 BC) - Symbol Placeholder

Who invented the Zero ?

- However, for easier calculations as we can see now Brahmagupta could have created a Zero out of nothingness(Śūnyatā).
  - Indians (~ 400 AD) – Symbol as DOT and Used in Arithmetic
  - The cultural and religious backdrop of Hinduism preaches everything starts from nothingness. So, he might as well give that 'nothing'-ness a symbol. And for Aryabhata, he gave the rules to use 0 in mathematical operations, not the number itself.

Abstract Math → Applied Math

Zero could have been invented for easier calculations and abstract mathematics.

There is a saying:

"Today's abstract math is tomorrow's applied math".

Documentary :

https://www.youtube.com/watch?v=9Y7gAzTMdMA

Before the Greeks

Neither Framework nor Functional Theories exists

Very useful collection of discoveries, but lack proof of mathematical relationships and its impact on physical phenomena

BBC Ancient Greece The Greatest Show on Earth - Democrats - 1/3

COSMOS – NEIL TYSON